

REAL-TIME SOCIAL DISTANCE ESTIMATION
USING YOLO CONVOLUTIONAL NEURAL
NETWORK AND EUCLIDEAN DISTANCE

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INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG (IIUC)
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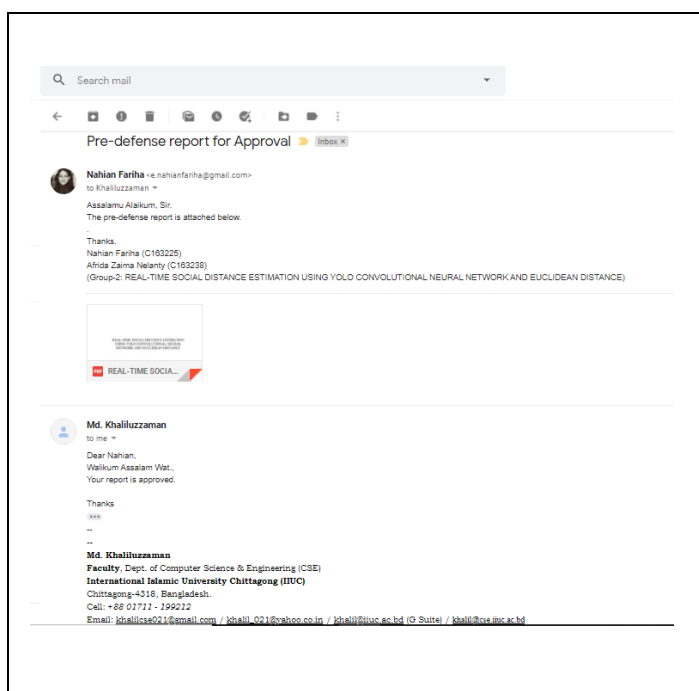
2020

SUPERVISOR'S DECLARATION

I am declaring that this thesis is read by me and with proper judgments, I think this thesis is good enough as a quality work and for the award of the degree of B. Sc. in Computer Science & Engineering.

23 February 2021

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DECLARATION

We articulate that this thesis is a result of our own hard work excluding a few quotations and compendiums that added knowledge to our research.

23 February 2021

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ABSTRACT

Social Distancing added a new interest in research with object detection. The main effective way of decreasing the infection being spread is to keep people distant from one another. Researchers are working with methods that can make social distancing more acceptable to people and estimation of it will make their life easier. The proposed method works with one of the latest CNN models 'YOLO' and Euclidean distance for estimation of social distance from top-down real-time videos. The method proposes the easiest way of social distance estimation compared to other CNN algorithms. Because the proposed algorithm is not only easy to understand but also gives a result that ensure pure speed and accuracy than previously used algorithms in object detection. This is proved on the result section by the percentage of tracking accuracy. Top-down videos and datasets collected from six different environments and situations are used to check the tracking accuracy. Tracking accuracy of 90% is then gained with this proposed method, which describes it's efficiency and splendid performance in object detection.

Keyword: Social Distancing, Person Detection, Distance Estimation, Python, YOLO, CNN

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Figure 4.2 Tracking Accuracy Result (in %)

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LIST OF ABBREVIATIONS

CNN	Convolutional Neural Network
YOLO	You Only Look Once
SPYDER	Scientific Python Development Environment
DNN	Deep Neural Network
FPS	Frame Per Second

CHAPTER I

INTRODUCTION

1.1 RESEARCH BACKGROUND

Social Distance, a way to lesser the percentage of getting infected with any viruses as well as Covid-19. In fact scientists also said that, may be in future we are going to fight with more deadly viruses like COVID-19. So, on a daily basis, Social Distancing will be a common approach for sensible people which is a useful method to control the spread of this type of virus. Researchers are working with social distance monitoring in a way so that social distancing can be more acceptable to people and monitoring of it will make their life easier specially for the people who have serious chronic medical conditions. Social Distance Monitoring requires real time object detection at first. Many deep CNN algorithms for real time object detection had been used already. But the method proposed here is a latest one.

In this research:

- the method is working with the latest Deep Convolutional Neural Network algorithm 'YOLO' and Euclidean distance
- social distance estimation with YOLO object detection algorithm is pre-trained with COCO dataset and here self-collected videos from six different environments are given as an input for checking a percentage of tracking accuracy
- tracking accuracy of 90% had gained with the research shown in the Result & Discussions chapter that proves the quality social distance estimation of the proposed method

1.2 PROBLEM STATEMENT

The main problem is public gathering where people are getting in physical contact and the spreading of virus has more chance. One way to slow the spread of viruses is social distancing. The super shops, hospitals, and institutions everyone is now concerned about maintaining social distance and monitoring of it. Social Distance Estimation is a new wanted research area of object detection. Old CNN algorithms are used in object detections. But the easier and latest the system, the better. A latest and an efficient algorithm is always a requirement in research.



Figure 1.1 Showing the results when person is in less contact [2]

1.3 MOTIVATION

Firstly, The recent pandemic created many research area not only in medical science but also in engineering. Object detection field is closely related to social distance

estimation which inspired this method including the research with the new algorithm of convolutional neural network.

Secondly, Social Distance is the new normal for all the people who are working out or in fact travelling. Institutions after the lockdown need to maintain social distance and in future with the attack of deadly viruses, situations requires social distance estimation. The motto is to get newer and easier way with more accuracy and speed. This motivated to research with social distance estimation.

1.4 OBJECTIVE OF RESEARCH

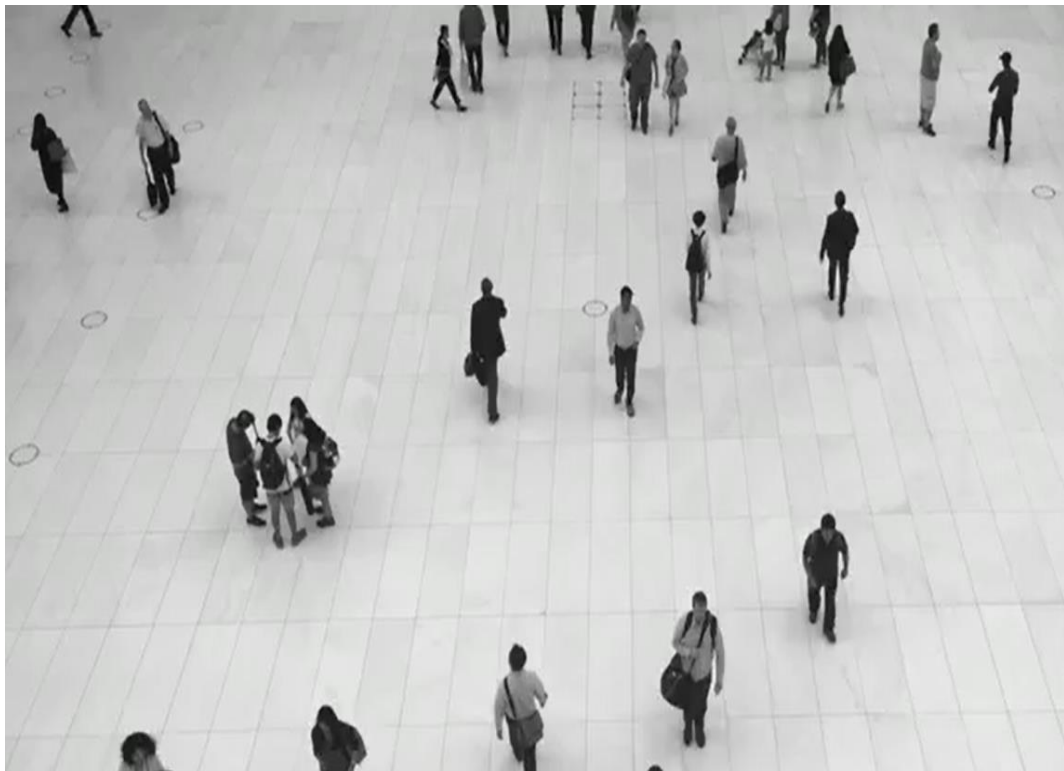


Figure 1.2 An image collected from a real-time video for estimation of social distance

The objective of this research is:

To estimate social distance in real-time and giving a tracking accuracy in percentage based on six different environments.

1.5 ORGANIZATION OF THE THESIS

Organizing of this thesis is a clear exhibition of proposed method that includes five chapters.

1. The first and introduction chapter presents the problem statement, the reason of choosing this area and objective in brief.
2. The second means the literature review chapter tells the scope and literature review in the area of deep learning based object detection, theoretical background, algorithms related to object detection.
3. The third chapter describes proposed methodology and point to point explanation of the proposed methodology including steps drawn in a diagram.
4. The fourth chapter discusses the aspired result with charts and samples and datasets .
5. The fifth chapter is included with future possibilities in our research area. The rest portion includes references, questionnaires and materials related to work.

CHAPTER II

LITERATURE REVIEW

2.1 INTRODUCTION

Social Distance Detection requires real time object detection at first. Detection of different classes of objects in deep learning shows fruitful results with accuracy. Many researchers utilized CNN algorithms for real-time person detection. This chapter is focused on literature review of proposed scenarios of before and present time related the method as they are the base key of this research.

2.2 SCOPE OF RESEARCH

The Covid-19 crisis possessed unwanted challenges for not only the retail sector or corporate world but also in everywhere public gatherings are common. In fact scientists also said that, maybe in future we are going to fight with more deadly viruses like COVID-19 .So, on a daily basis, Social Distancing will be a common approach for sensible people and will control the spreading. So easier ways of detecting social distance and new ways of maintaining or monitoring social distance is the new popular scope of research.

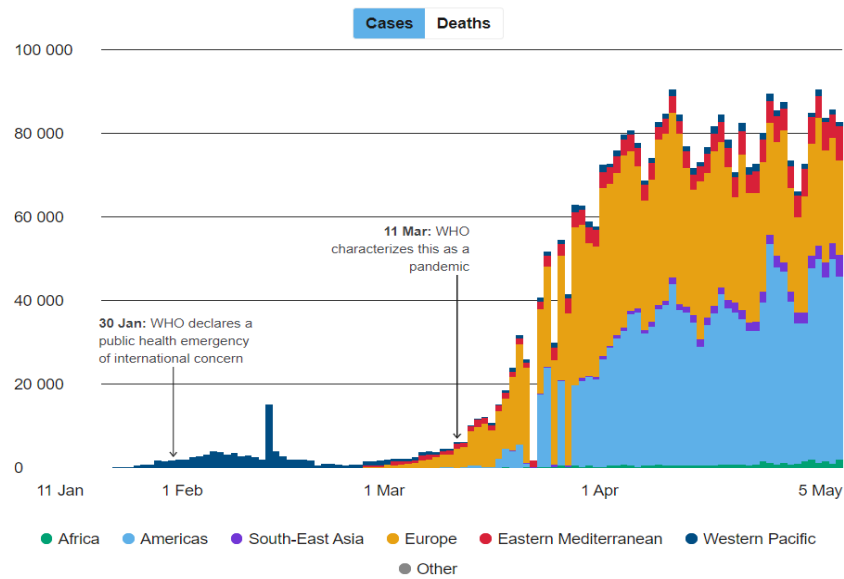


Figure 2.1 Daily reported cases and deaths of COVID-19[3]

Researchers are working with this method so that social distancing can be more acceptable to people and monitoring of it will make their life easier. Real time object detection algorithms made social distance detection easier. Some of the algorithms of real time object detection include datasets trained in advance. CNN used in real-time object detection provides enriched research scope.

2.3 LITERATURE REVIEW

This literature review section includes two parts. First part includes theoretical background relates object detection The next section includes the theoretical background with IOT based, statistics based and object detection based social distance estimation.

2.4 THEORETICAL BACKGROUND

Object Detection is popular and so updated models are proposed quite often. Fast Region-Based Convolutional Network had quality of high detection and no disk was required for feature caching. Both python and C++ were used here. Fast R-CNN usually trained on the ImageNet and consists of ROI pooling layer, softmax layer branch, bounding box regression branch etc.

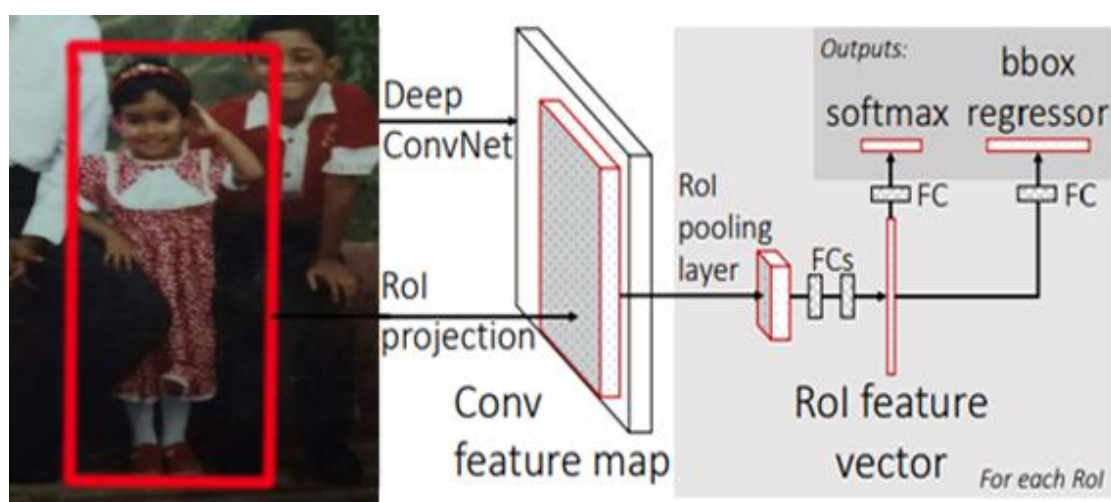


Figure 2.2 Architecture of Fast Region-Based Convolutional Network

Faster R-CNN follows features like full frame convolution, high quality generating and works in a cost effective manner. Another way HOG offers more understandable features. HOG follows ROI. ROI means regions of interest which is a gradient orientation in localized portions of a frame.

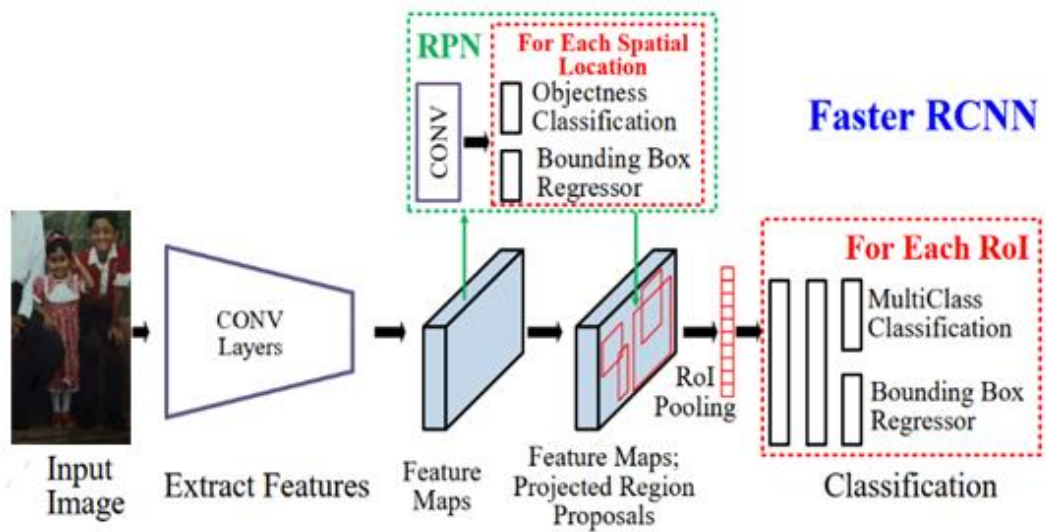


Figure 2.3 Architecture of Faster R-CNN

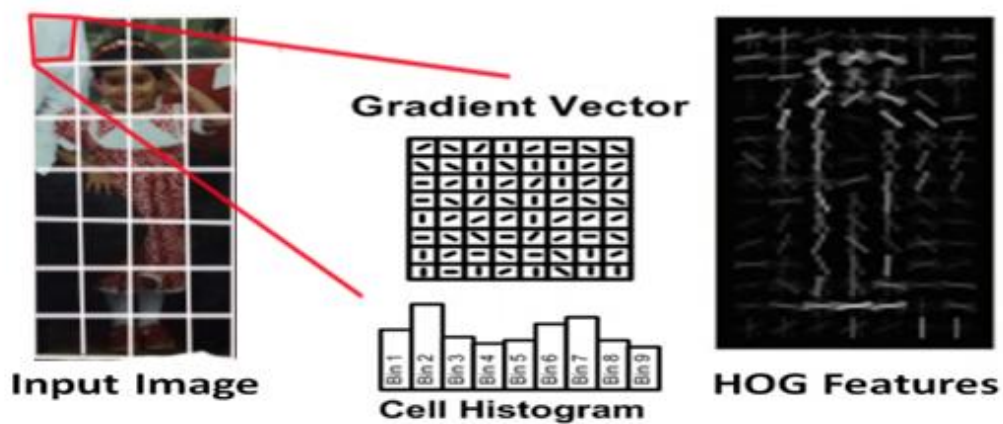


Figure 2.4 Architecture of histogram of Oriented Gradients (HOG)

R-CNN have the ability to train a high capacity model and it localizes objects with a deep network and needs only a small quantity of data, achieves quality accuracy and scales classes. For example: hashing.

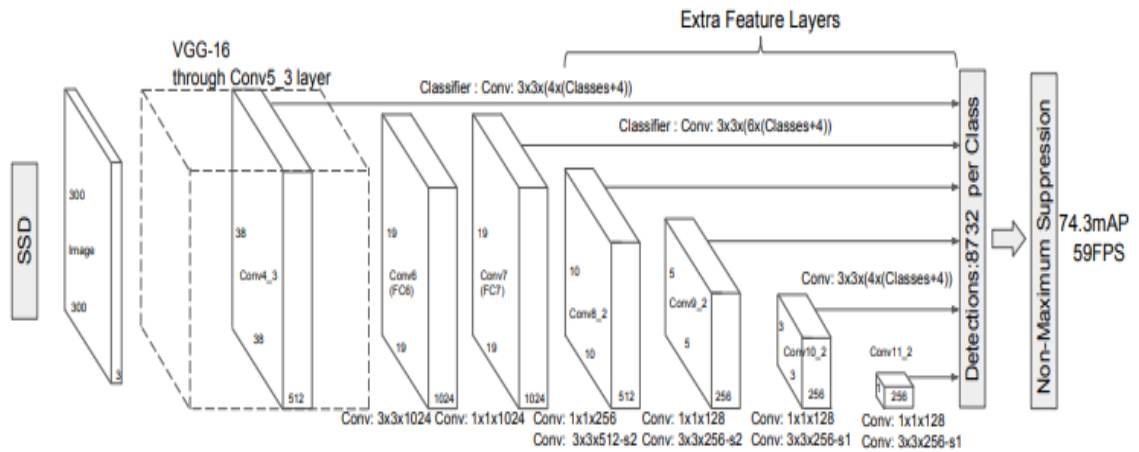


Figure 2.5 Architecture of Single Shot Detector (SSD) [1]

Handling various sizes of objects by predicting from different features are done by Single Shot Detector.

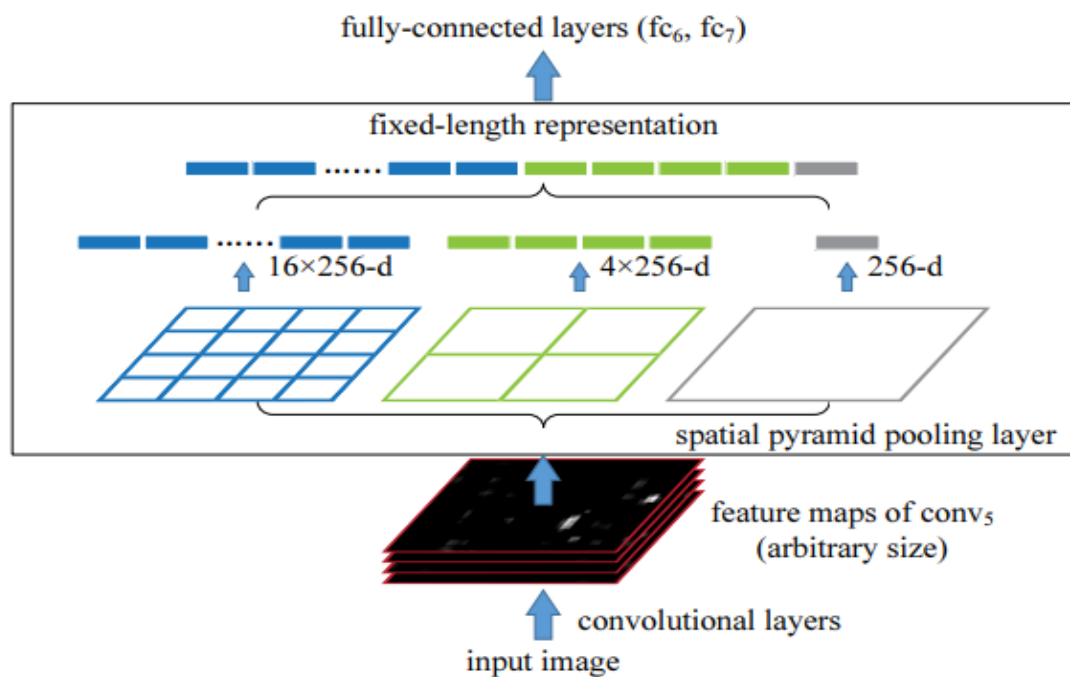


Figure 2.6 Architecture of Spatial Pyramid Pooling (SPP-net) [1]

Spatial Pyramid Pooling works in a way that can avoid repeat in case of computing features that are convolutional, pool features, generates fixed-length representation to train detectors.

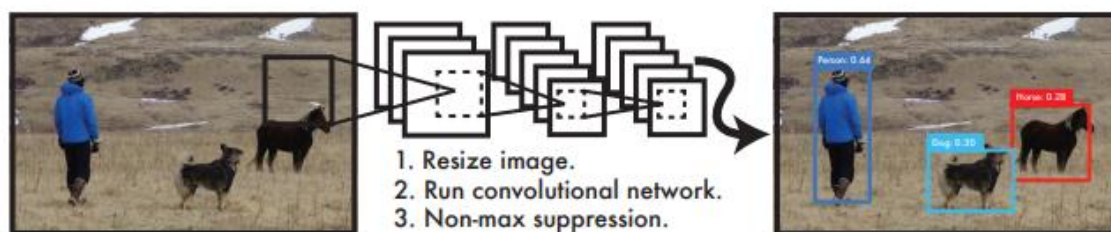


Figure 2.7 YOLO (You Only Look Once) [1]

YOLO follows a fast manner, achieves double the mAP than others in fact some model of YOLO processes images at 45 frames per second. YOLOv3 is extremely fast and accurate and about 4x faster.

2.4.1 Researches related to Social Distance

Object detection in real time is popular. But previously IOT based research in social distance detection is used and there was statistics based research as well. There was a scientific indicator named reproduction ratio was used for social distancing criteria. This reproduction ratio was the way of giving information about the average of people getting infected going closer to another virus infected person in a certain society. If the reproduction ratio is lesser than one then it was estimated that lesser than one person will get infected and if reproduction ratio shows otherwise the case gets opposite. This reproduction ratio showed rate as 2-2.5 in the case of corona which is higher than any other pandemic situation of past decades. Another social distance parameter was an SIR model that used to give result on the basis of recovered and infected person.

Then comes tracking based methods in social distance. A program named "Aarogya Setu" was introduced and developed in India for contact tracking of infected persons came in a mobile app and installed by 100million people of India. It includes features like user status, self-assessment, updates of the disease etc. IOT based researchers introduced drones and surveillance cameras to some law enforcement agencies for over populated rallies. Though AI based researches needs practical and mental work,

these methods are most useful from the beginning of object detection. Computer understands the visual data from real-time videos or frames by the help of computer vision and deep learning. Also identification and classification of objects are part of it and as a result such features are used in CCTV and smart cameras. A study shows to use MobileNet for lightweight detecting, but the problem here was less accuracy than other detecting models. Previously, object detection models such as RCNN, fast RCNN, faster RCNN, Single Shot MultiBox Detector, YOLOv1-v4 are tested on PASCAL Visual Object Classes data sets and MS COCO under similar conditions. The performance depends on their backbone architecture, frame size and model depth too.

Considering those research gaps, the study proposes a new approach that is tested on a large dataset in advance and ensures good percentage of accuracy and speed. The proposed study estimation will work in process of quick detection with 30FPS.

2.5 SUMMARY

This chapter contains two sections: theory and work-based literature review. In the first section, we have mentioned the basic deep learning object detection algorithm, OpenCV library, YOLO COCO we studied and in the next section we mentioned previous research works including human tracking, statistics and IOT-based research and other object detection-based research. All this research work has two things in common, they are of real time and directly or indirectly related to human tracking and social distancing.

CHAPTER III

METHODOLOGY

3.1 INTRODUCTION

A method of systematically solving a research problem is called Research Methodology. It aims to provide a research work plan, training in building models, tools and procedures that go along with the solution to the problem. This chapter is about the proposed methodology of social distance estimation, presenting a deep learning-based social distance estimation model for top-down footages.

3.2 METHODOLOGY

The proposed method is evaluated on real time video, from the video the desired frames are gained. Yolo object detector detects object where on the proposed method only person is detected from the desired frame. Then the formula of Euclidean distance helps to determine the distance in pixels using each one's centroid information. If violation occurs, the bounding box's color is shown in red bounding box and if not then the green bounding boxes are generated for people in safe distance. Thus the social distance estimation from the number of violating can be estimated. The total number of detected person and correctly detected shows the estimation perfectly, which will be discussed in the fourth chapter briefly.

The Workflow diagram of proposed method:

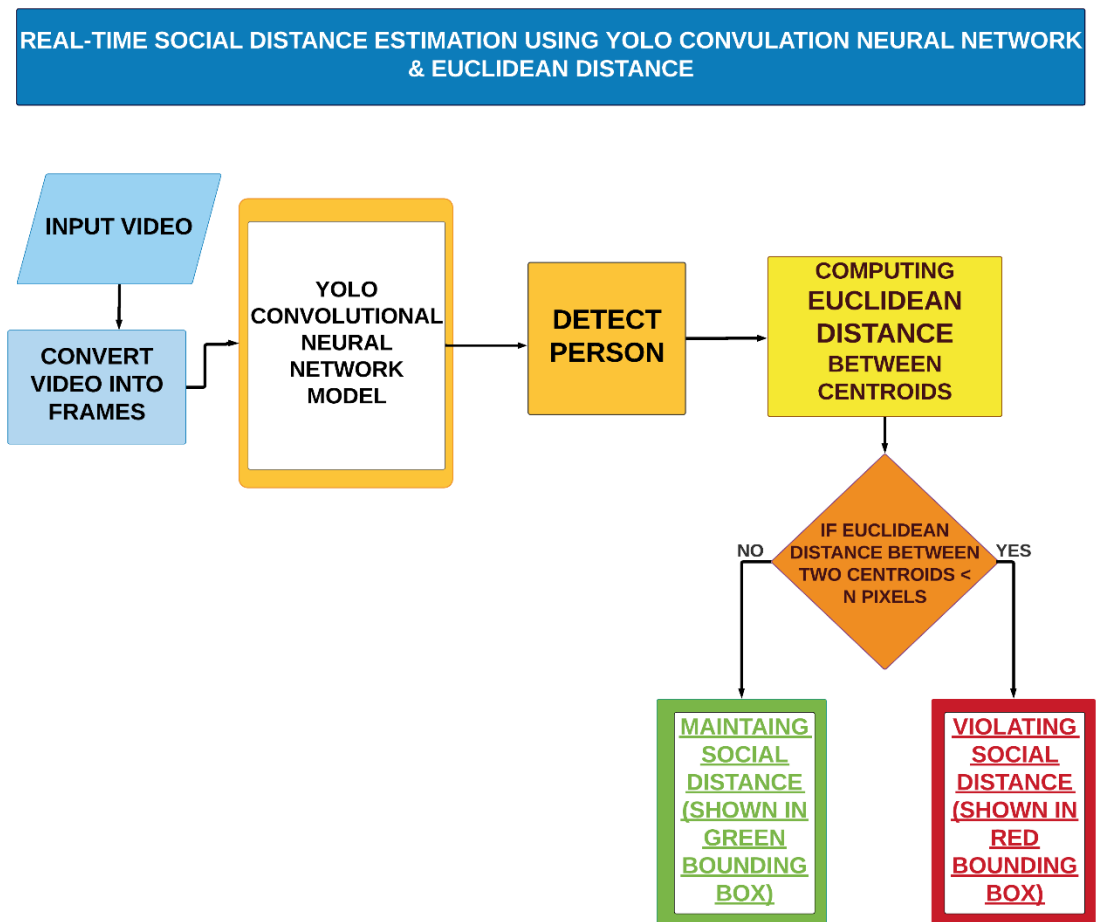


Figure 3.1 Work flow diagram of the proposed system (N=50)

i. INPUT VIDEO:

Sample videos are collected from different environments that are in different conditions in the form of mp4 file. Six sample videos are kept in the same directory where python code is present to run the python interpreter.

ii. CONVERT VIDEO INTO FRAMES

Frame is used from video file. Pre-Process images taken by a video to detect the components of the image and classify them by using python. Most of the video

formats to number of frames work on python 3 and OpenCV which supports mp4 file to convert video into frames.

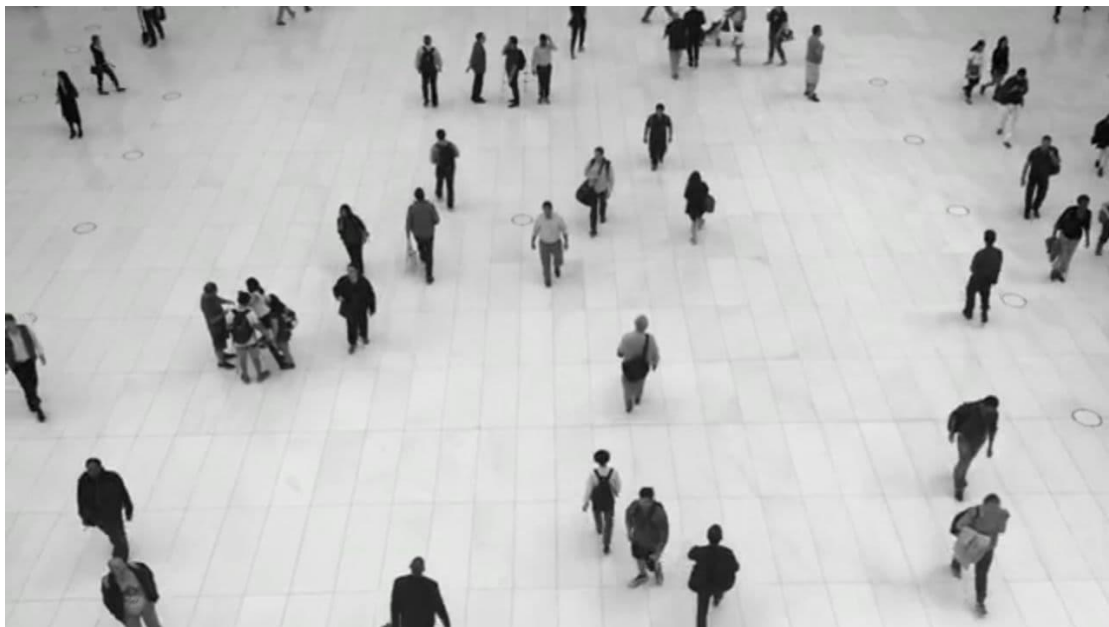


Figure 3.2 Frame Collected from Input Video

A method was constructed that can perform OpenCV with YOLO requires frame rate, box scores, person centroids and person detection confidences a frame will be showing results according to it. A blob is created for extracting the dimensions from the frame which will pass through CNN and outputs prediction.

iii. YOLO CONVOLUTION NEURAL NETWORK MODEL

After successfully converting video into frames, YOLO based CNN model was applied where pre-trained weights and class names used as YOLO is compatible with real time person detection module.

The approach involves a single deep neural network called DarkNet which splits the input into grid cells that directly predicts a bounding box and its confidence scores. Here, YOLOv3-608 with DarkNet-53 backbone working on pre-trained COCO dataset. DarkNet-53 contains 53 convolution layers which followed by batch

normalization and leaky relu activation, pooling is unavailable as it strides two down-sample feature maps. Sigmoid function used for objectness score and class confidence because we detect only person. COCO has recognition, instances over 1.5 million object, 80 object categories, 91 stuff categories and 5 captions per image. A fully CNN architecture generating 1x1 kernel with 3 different sizes and 3 different networks. A blob is constructed so that object detection can be performed with YOLO and OpenCV. Bounding boxes, Person centroids and its confidence scores initialized for inference which helps looping over each of the layer outputs.

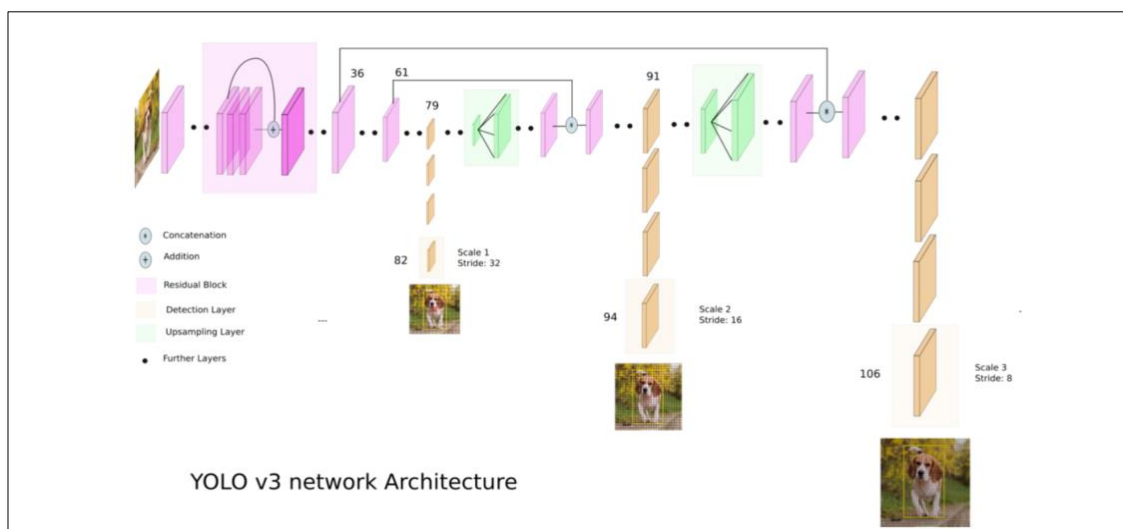


Figure 3.3 The figure is showing YOLOv3 network architecture [25]

It takes (608x608) image by pre trained model with trained DarkNet-53 backbone volumes (19x19x425) detections at three scales. Output shape at each scale performs (batch size x number of anchor boxes x grid size x 85 dimensions).

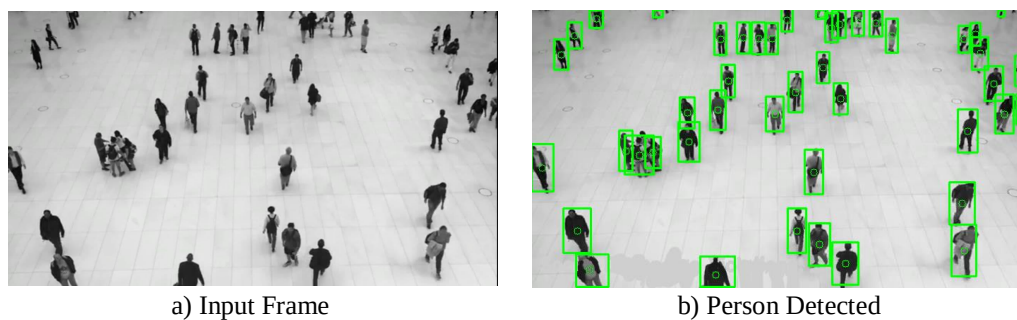


Figure 3.4 Experimental Result of: a) input frame, b) person detect

Extracting DarkNet class id and probability of person are computed. $425 = 5 \times 85$ because each cell predicts 5 boxes corresponded with 5 anchor boxes. Fig 3.4 is showing how Yolo neural network extracted features from images.

iv. DETECT PERSON

Processing the video we got some frames in different conditions and different places .There are many objects seen in the video but the person class is only detected as it is the main priority here.

Yolov3 trained on COCO dataset is used for object detection in our proposed method. CNN architecture structured with weights and class names as it is compatible with deep neural network for real time detection. Sigmoid function used for objectness score and class confidence because we detect only person. For coco dataset this is true that this object fits in to one class, not belong to other classes.



a) Input Frame

b) Person Detected Frame

Figure 3.5 a) input frame, b) person detected frame

We only consumed value that only detect people from 80 other classes.

$85 = 5 + 80$; class number = 80 (yolo) ; (pc, bx, by,bh, bw) = 5

OpenCV and YOLO combindly worked here as person detection from video and classify them only one class .Drawing bounding box around persons and finding the centroids of the box is also a feature of YOLO.

v. COMPUTE EUCLIDEAN DISTANCE BETWEEN CENTROIDS

From the previous step, after the person class detected now computation between centroids is needed .Euclidean distance estimation algorithm computes bounding boxes, thus representing the true detection of the person. For measuring the Euclidean distance at first we need object tracking. A scenario is created for every frame as following for object tracking:

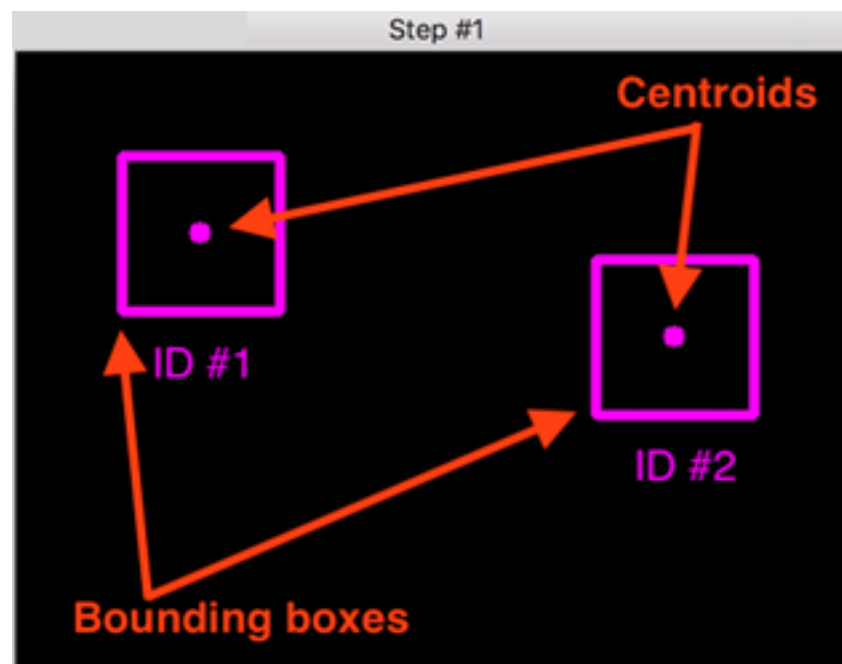


Figure 3.6 Step -1

Step #1 is showing the first frame. Simply , a centroid is the center of a bounding box and two ID (ID #1 , ID #2) names are given to those bounding box.

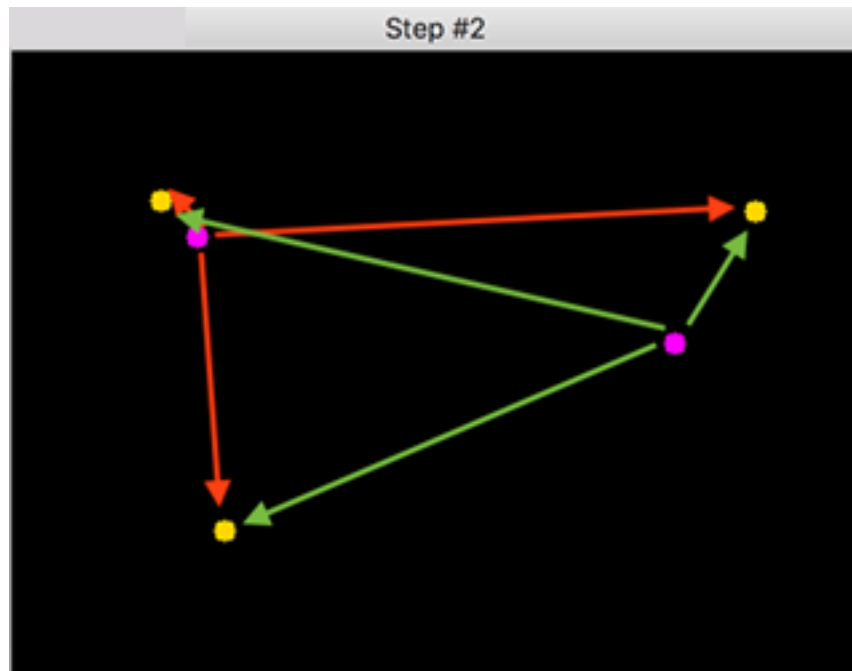


Figure 3.7 Step-2

Step #2 is the next video frame where the purple one is shown as old centroid and the yellow one is showing new centroid and the single yellow one is showing as a new person detected. Lines are drawn from every old centroid to new centroid to know the movement.

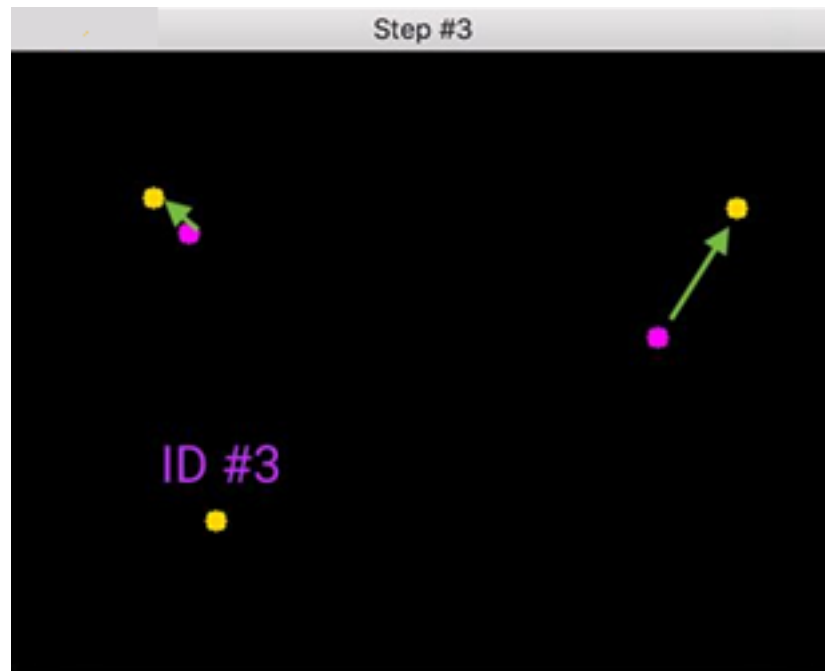


Figure 3.8 Step-3

Step #3 is showing the new detected person as ID name ID #3.

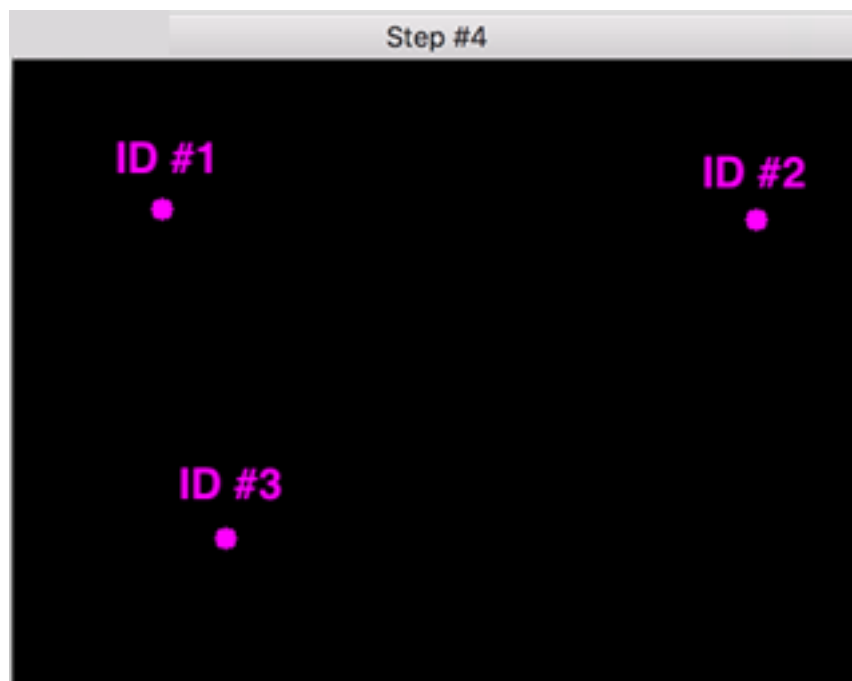


Figure 3.9 Step-4

Step #4 is showing three ID names. If one of those ID cannot be found for consequent 40frames, the ID will be erased.

Then distance from Camera to person is detected:

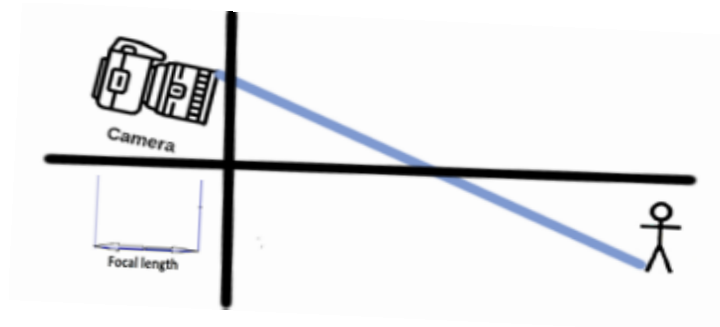


Figure 3.10 Camera to person detection (1)

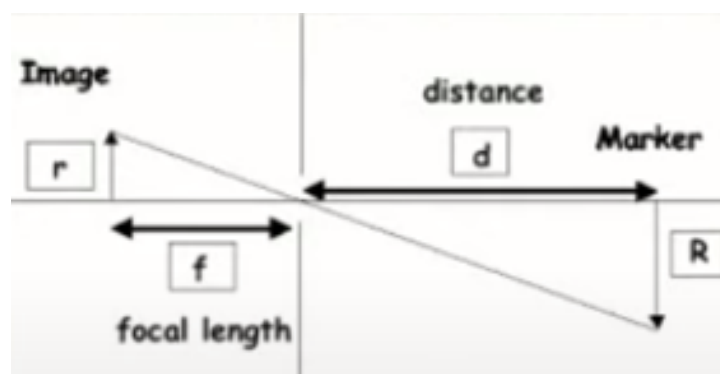


Figure 3.11 Camera to person detection (2)

For camera to person detection, Focal length is measured by the following formula:

$F = (P \cdot D) / W$; where F = Focal Length, P = Pixel the object is covering in the photo, D = Distance from object to camera's focal length, W = Wide of the object

$D' = (W \cdot F) / P$; where D' = Camera to person distance, F = Focal Length

Then, after measuring camera to person distance, Person to person distance is measured by Euclidean distance formula:

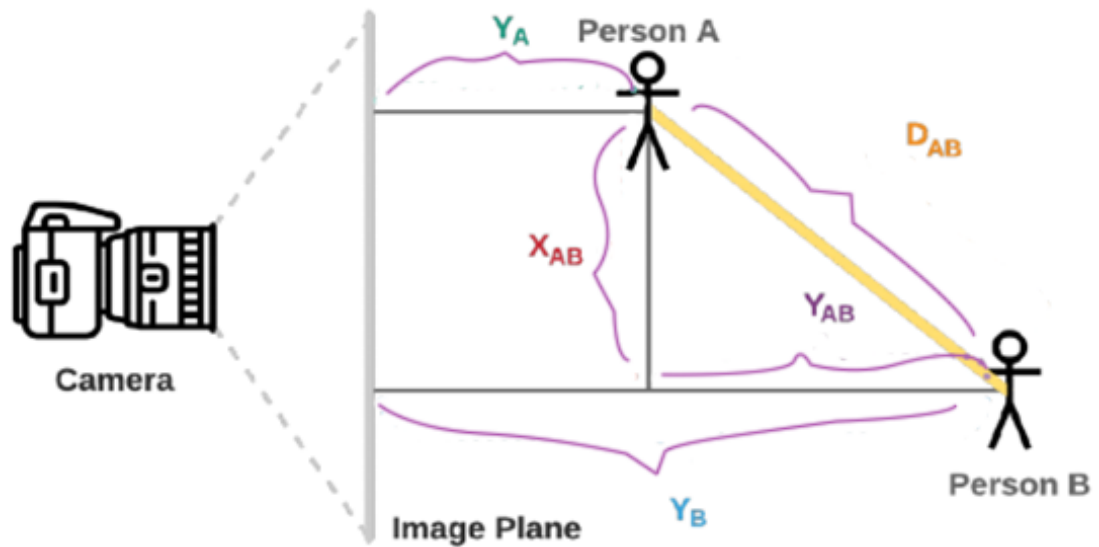


Figure 3.12 Person-to-person distance

$$D = \sqrt{(Y_B - Y_A)^2 + (X_B - X_A)^2}$$

Figure 3.13 Euclidean distance formula

Then gained distance between two centroids for people were result to the next step.

vi. IF DISTANCE BETWEEN TWO CENTROIDS < N PIXELS

1. Here, cut-off distance for social distancing is 50 pixels. That means, N=50.
2. If the Euclidean distance between two centroids < N pixels, the bounding box will be shown in red and if the Euclidean distance between two centroids > = N pixels, then the bounding box will be shown in green box.

3. Finally, Total number of social distancing violations will be displayed on the output frame.
4. Violation number will help to estimate the social distance.

3.3 SUMMARY

This chapter includes the whole methodology and research activity in brief. The next chapter shows the result of this proposed method with necessary data, charts and calculations.

Social Distance Detection parameter is applied to get the result. This whole process is a recursive process. A detection function takes arguments and extract the dimension of the frame. Because the distance between objects will be measured with respect to other objects present in that real time video. Our cut-off distance for social distancing is 50 pixels. If the distance between two centroids is less than that than the output will be in red box and if not then the output will be in green box. In our proposed method YOLO v3 calculates more accurate result for object confidence (for person) and class predictions are predicted through logistic regression.

CHAPTER IV

RESULTS AND DISCUSSION

4.1 INTRODUCTION

In this chapter, the overall experiments analyses and results with necessary figures, charts and calculation will be covered. At the end of this chapter, a tracking accuracy will be shown that is gained from sample video frames of six different type or environment.

4.2 RESULTS

The detailed experiment review of the model was described in methodology chapter of proposed method. Six different types of situations and environments were taken as input data to get an acceptable tracking accuracy in percentage.

Footage environments or situations are of following characteristics:

- 1) Frame 1: Good Quality
- 2) Frame 2: Fuzzy
- 3) Frame 3: Daylight
- 4) Frame 4: Nightlight
- 5) Frame 5: Less Populated
- 6) Frame 6: Crowded

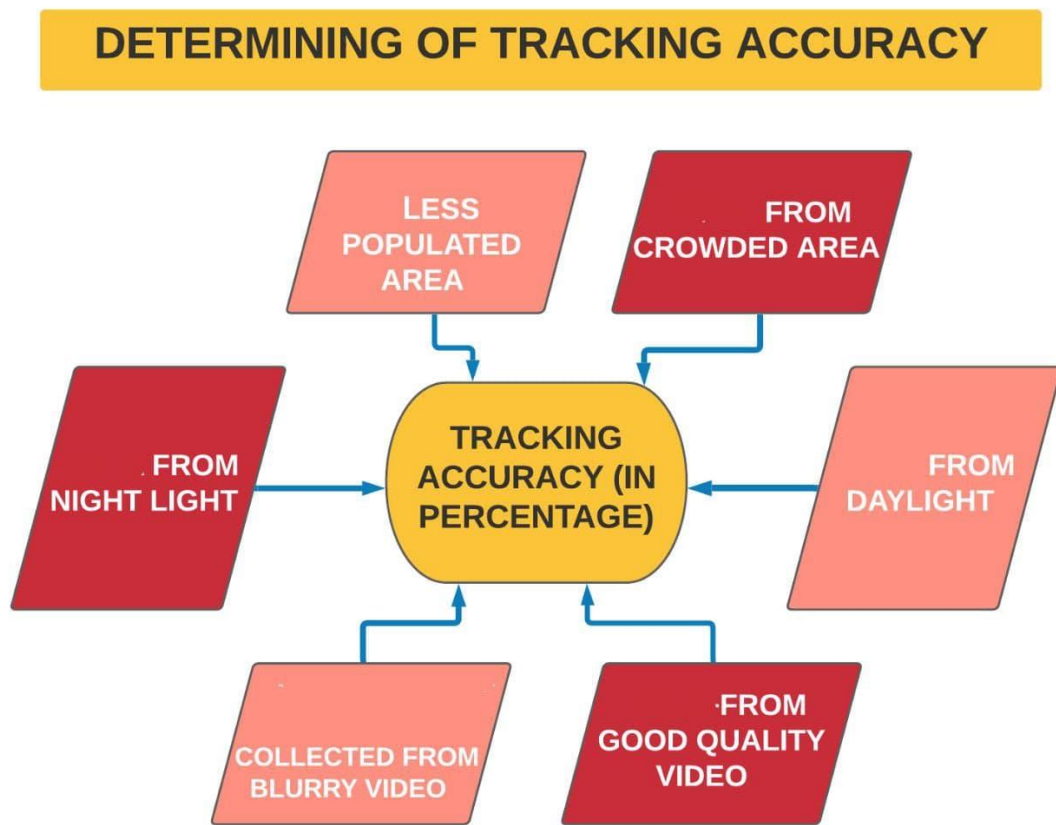


Figure 4.1 Tracking Accuracy (in percentage) measurement process

Datasets from different environment and situations prove the model's tracking accuracy and efficiency in different environments. Here, six environments will show six different of percentage of accuracy depending on the footage or situation. The result section contains five frames for each video and then their average percentage. Lastly, the average percentage will be shown as the whole tracking accuracy of proposed method. The result section contains five frames for each video.

4.2.1 Environment from Good Quality Footage

This footage has high resolution feature and clarity of people walking.

Table 4.1 **Frame from Good Quality Footage**

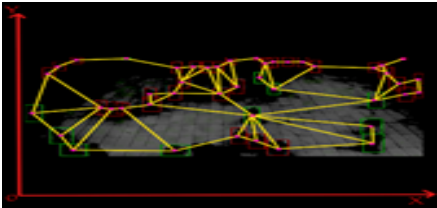
Step-1. Frame from Input Video 	Step-2. Person Detected 
Step-3. Euclidean distance measurement 	Step-4. Social Distance Violance estimated 

Table 4.2 **Frame from input video**

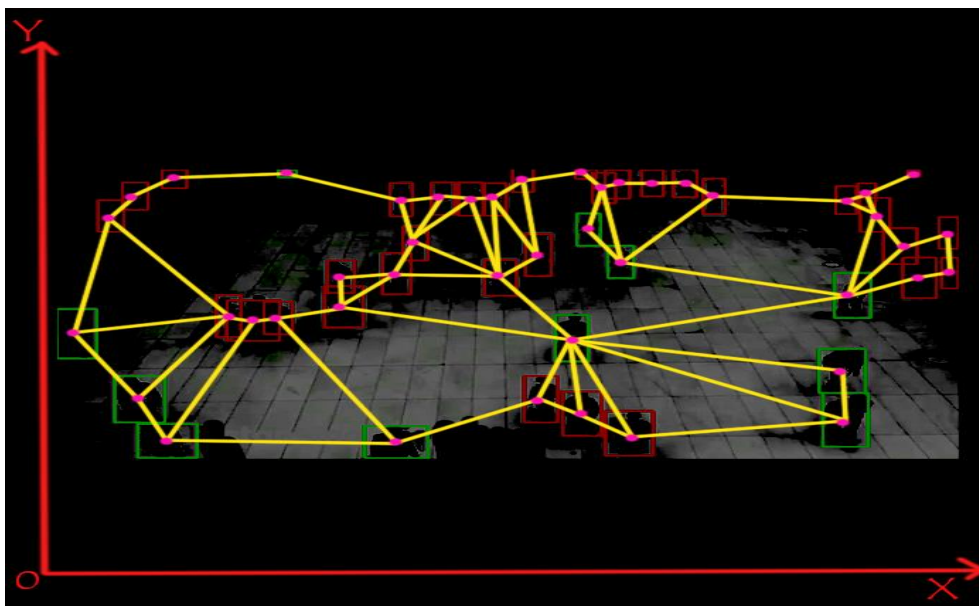
Frame:1, Step-1. Frame from Input Video

Total Person : 45

Table 4.3 **Person Detected**

Frame:1, Step-2. Person Detected : 45

Table 4.4 Person detection and Euclidean distance measurement



Step-3. Frame:1, Euclidean distance measurement

Table 4.5 Social Distance Violence estimated



Step-4. Frame:1, Correctly Detected : 45

Table 4.6 More Sample Frames from the same video

 Social Distancing Violations: 27 Frame: 2 Total Person: 38 Correctly Detected: 38	 Social Distancing Violations: 27 Frame: 3 Total Person: 40 Correctly Detected: 39
 Social Distancing Violations: 29 Frame: 4 Total Person: 40 Correctly Detected: 39	 Social Distancing Violations: 29 Frame: 5 Total Person: 40 Correctly Detected: 39

We calculated Tracking Accuracy for frame in percentage by multiplying the number of correctly detected with 100 and dividing it by the total person.

That means, Tracking Accuracy for frame in percentage = No. of Correctly Detected*100 / No. of Total person

Frame 1= 100% , Frame 2= 98%, Frame 3= 98% , Frame 4= 98% , Frame 5= 100%

Tracking Accuracy of whole environment in percentage = The Sum of Tracking Accuracy for frames in percentage / Total Frame = $100+98+98+98+100/5=99\%$ (approximate)

Tracking Accuracy gained from Good Quality Footage: 99%

4.2.2 Environment from Fuzzy Footage

Table 4.7 **Frame from Fuzzy Footage**

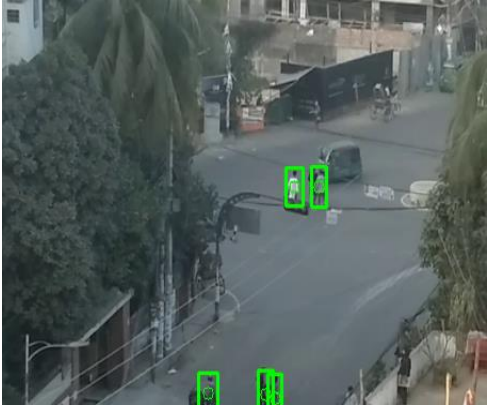
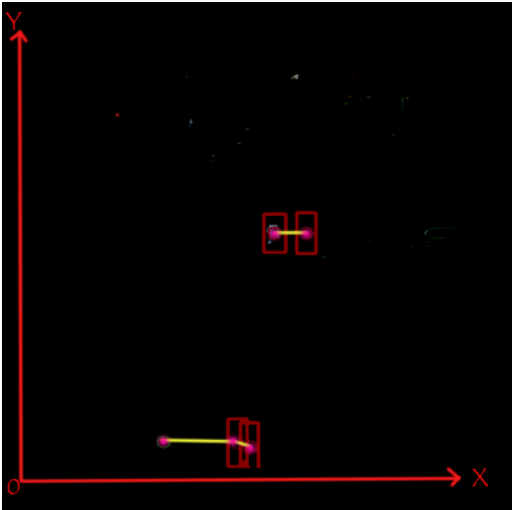
Step-1. Frame from Input Video 	Step-2. Person detected 
Step-3. Euclidean distance measurement 	Step-4. Social Distance Violance estimated 

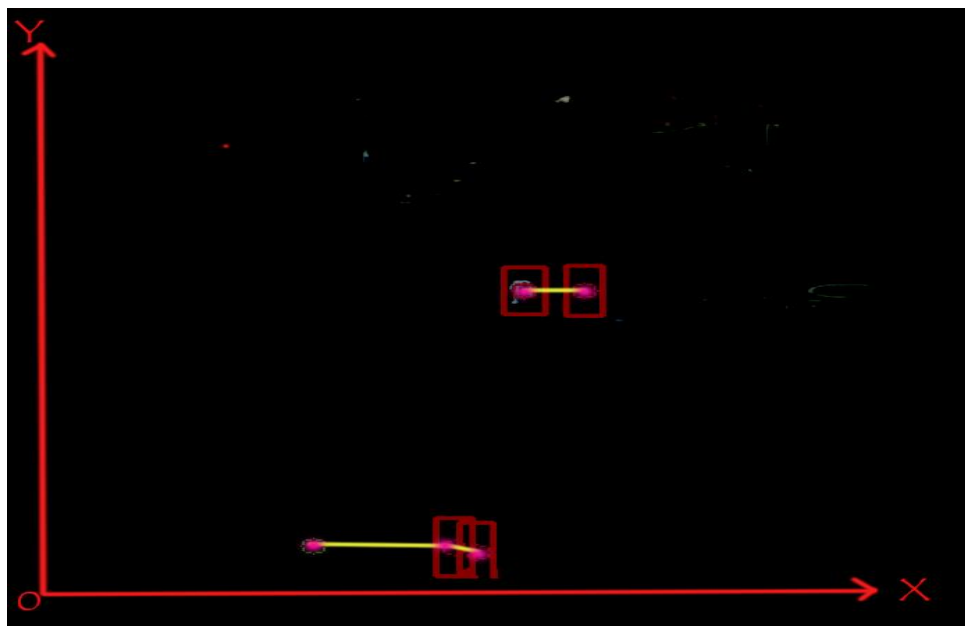
Table 4.8 **Frame from Input Video**

Frame:1, Step:1. Total Person: 07

Table 4.9 **Person Detected**

Frame:1, Step 2. Person Detected : 05

Table 4.10 Person detection and Euclidean distance measurement



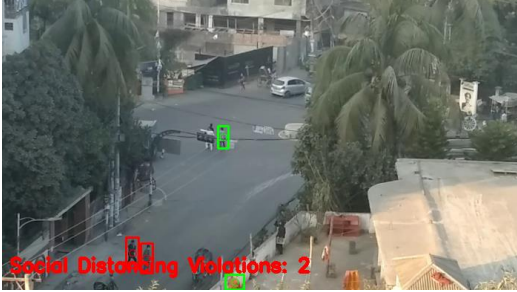
Frame:1, Step-3. Euclidean Distance Measurement

Table 4.11 Social Distance Violence estimated



Frame:1, Step-4. Detected Person: 05

Table 4.12 More Sample Frames from the same video

 Frame:2 Total Person: 08 Correctly Detected: 05	 Frame:3 Total Person: 08 Correctly Detected: 05
 Frame:4 Total Person: 08 Correctly Detected: 05	 Frame:5 Total Person: 07 Correctly Detected: 04

Here,

Tracking Accuracy for frame in percentage = $\frac{\text{No. of Correctly Detected} \times 100}{\text{No. of Total person}}$

Frame 1= 63% , Frame 2= 63%, Frame 3= 63% , Frame 4= 57% , Frame 5= 72%

Tracking Accuracy of whole environment in percentage = $\frac{\text{The Sum of Tracking Accuracy for frames in percentage}}{\text{Total Frame}} = \frac{63+63+63+57+72}{5} = 64\%$
(approximate)

Tracking Accuracy gained from Fuzzy Footage: 64%

4.2.3 Environment from Daylight Footage

Table 4.13

Frame from Daylight Footage

Step-1. Frame from Input Video 	Step-2. Person detected 
Step-3. Euclidean distance measurement 	Step-4. Social Distance Violance estimated 

Table 4.14 **Frame from Input Video**

Frame:1 , Step:1. Total Person: 16

Table 4.15 **Person Detection**

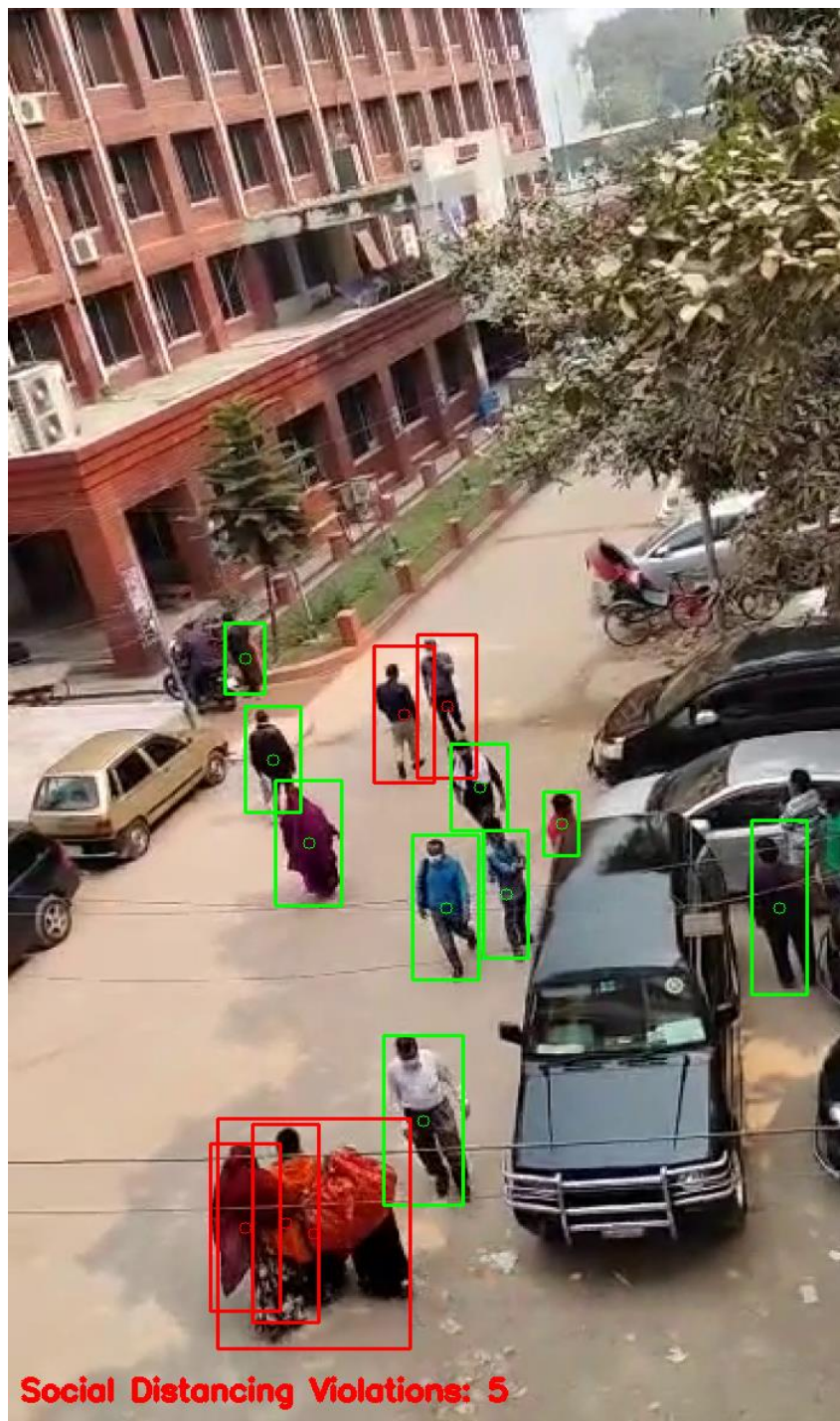
Frame:1, Step-2. Person detected : 14

Table 4.16 Euclidean distance measurement



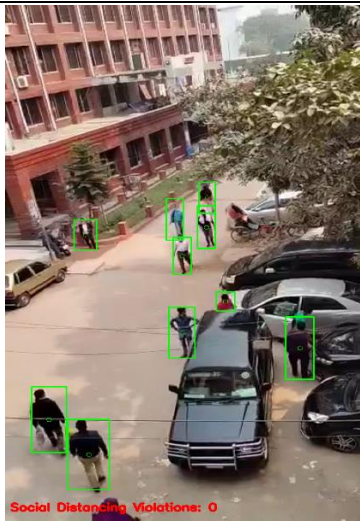
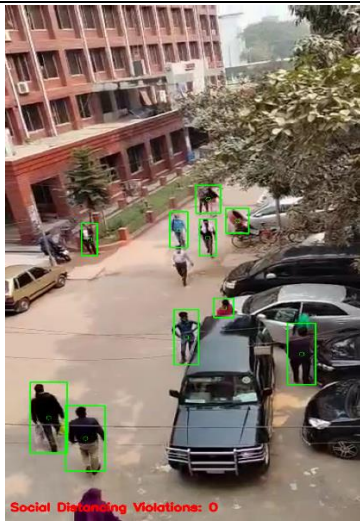
Frame:1, Step-3. Euclidean Distance Measurement

Table 4.17 Social Distance Violence estimated



Frame:1, Step-4. Correctly Detected :14

Table 4.18 **More Sample Frames from the same video**

 Frame:2 Total Person: 12 Correctly Detected: 10	 Frame:3 Total Person: 12 Correctly Detected: 10
 Frame:4 Total Person: 13 Correctly Detected: 11	 Frame:5 Total Person: 16 Correctly Detected: 14

Here,

Tracking Accuracy for frame in percentage = No. of Correctly Detected*100 / No. of Total person

Frame 1= 88% , Frame 2= 84%, Frame 3= 93% , Frame 4= 85% , Frame 5= 84%

Tracking Accuracy of whole environment in percentage = The Sum of Tracking Accuracy for frames in percentage / Total Frame = $88+84+93+85+84/5=87\%$ (approximate)

Tracking Accuracy gained from Daylight Footage: 87%

4.2.4 Environment from Nightlight Footage

Table 4.19 **Frame from Nightlight Fuzzy Footage**

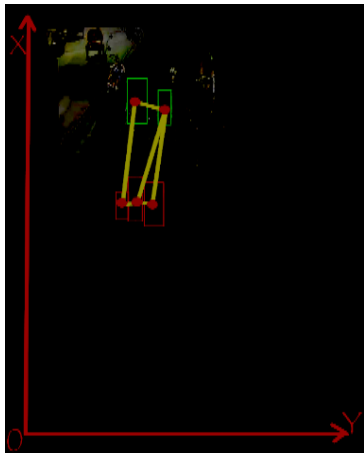
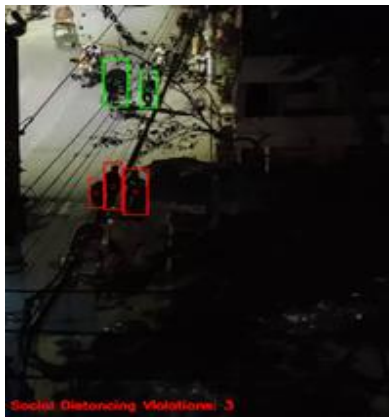
Step-1. Frame from Input Video 	Step-2. Person detected 
Step-3. Euclidean distance measurement 	Step-4. Social Distance Violance estimated 

Table 4.20 **Frame from Input Video**

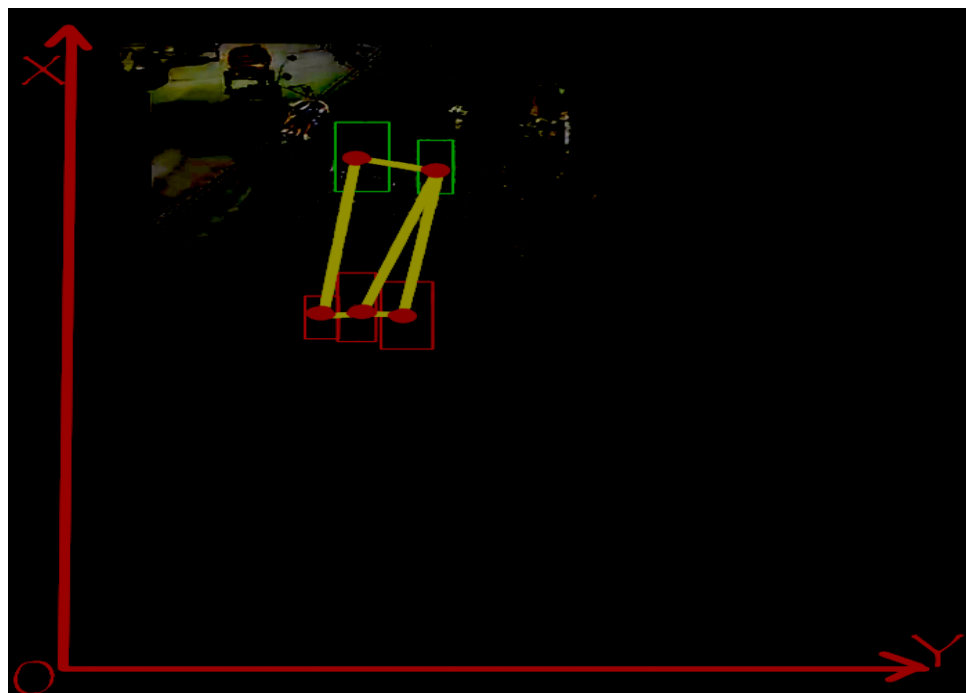
Frame:1, Step-1. Total Person: 05

Table 4.21 **Person Detection**

Frame:1, Step-2. Person Detected: 05

Table 4.22

Person detection and Euclidean distance measurement



Frame:1, Step-3. Euclidean Distance measurement

Table 4.23

Social Distance Violence estimated



Frame:1, Step-4. Detected Person : 05

Table 4.24 **More Sample Frames from the same video**

 Frame:2 Total Person: 05 Correctly Detected: 04	 Frame:3 Total Person: 05 Correctly Detected: 04
 Frame:4 Total Person: 05 Correctly Detected: 05	 Frame:5 Total Person: 05 Correctly Detected: 05

Here,

Tracking Accuracy for frame in percentage = No. of Correctly Detected*100 / No. of Total person

Frame 1= 100% , Frame 2= 80%, Frame 3= 80% , Frame 4= 80% , Frame 5= 100%

Tracking Accuracy of whole environment in percentage = The Sum of Tracking Accuracy for frames in percentage / Total Frame = $100+80+80+80+100/5 = 92\%$ (approximate)

Tracking Accuracy gained from Nightlight Footage: 92%

4.2.5 Environment from Less Populated Footage

Table 4.25 Frame from less populated area Footage

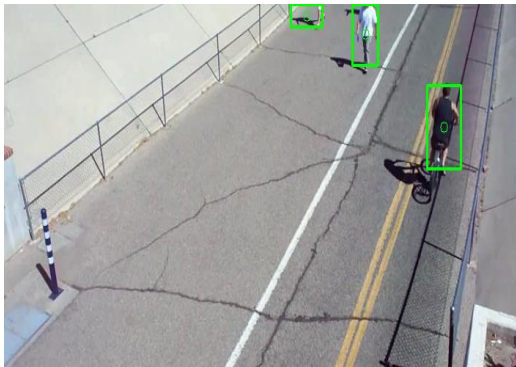
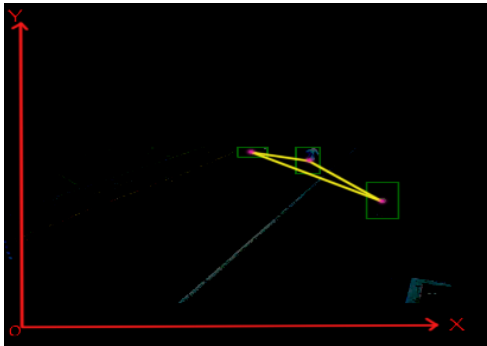
Step-1. Frame from Input Video 	Step-2. Person detected 
Step-3. Euclidean distance measurement 	Step-4. Social Distance Violance estimated 

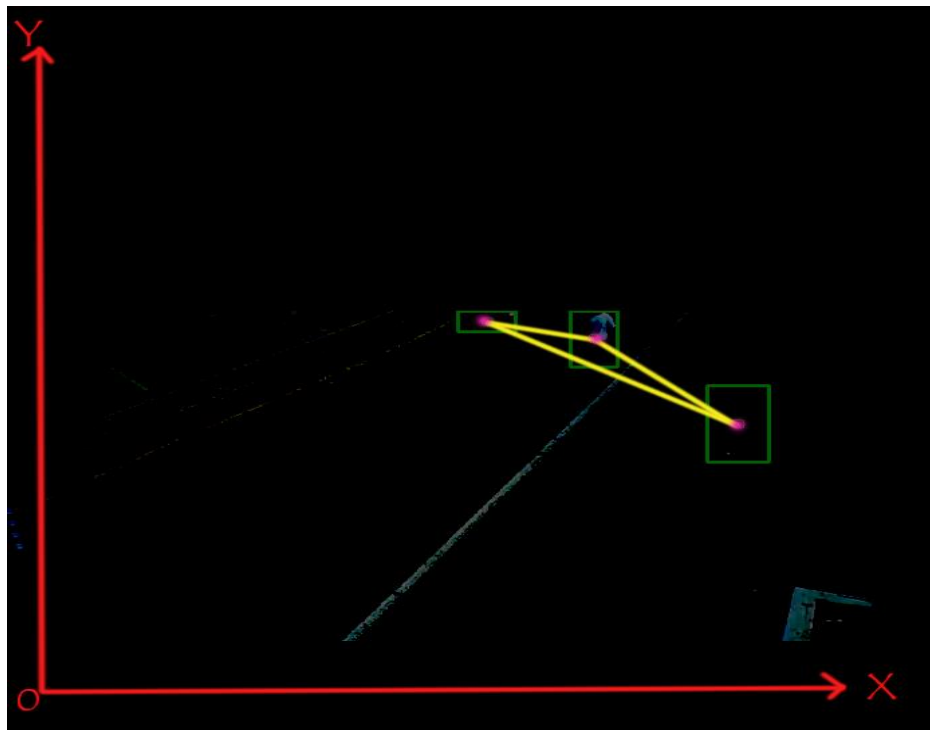
Table 4.26 **Frame from Input Video**

Frame:1, Step-1. Total Person : 03

Table 4.27 **Person Detection**

Frame:1, Step-2. Person Detected: 03

Table 4.28 Person detection and Euclidean distance measurement



Frame:1, Step-3. Applying Euclidean Distance formula

Table 4.29 Social Distance Violance estimated



Frame:1, Step-4. Detected Person : 03

Table 4.30 More Sample Frames from the same video

 Frame: 2 Total Person: 03 Correctly Detected: 03	 Frame: 3 Total Person: 03 Correctly Detected: 03
 Frame: 4 Total Person: 03 Correctly Detected: 03	 Frame: 5 Total Person: 03 Correctly Detected: 03

Here,

Tracking Accuracy for frame in percentage = No. of Correctly Detected*100 / No. of Total person

Frame 1= 100% , Frame 2= 100%, Frame 3= 100% , Frame 4= 100% , Frame 5= 100%

Tracking Accuracy of whole environment in percentage = The Sum of Tracking Accuracy for frames in percentage / Total Frame = $100+100+100+100+100/5=100\%$ (approximate)

Tracking Accuracy gained from Less Populated Footage: 100%

4.2.6 Environment from Crowded Footage

Table 4.31 Frame from crowded area Footage

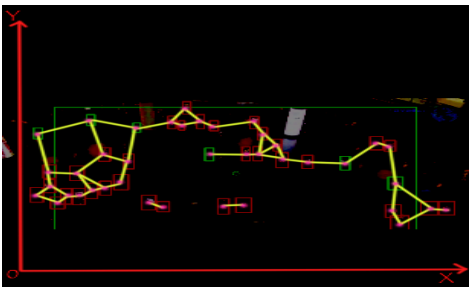
Step-1. Frame from Input Video 	Step-2. Euclidean distance measurement 
Step-3. Euclidean distance measurement 	Step-4. Social Distance Violance estimated 

Table 4.32 **Frame from Input Video**

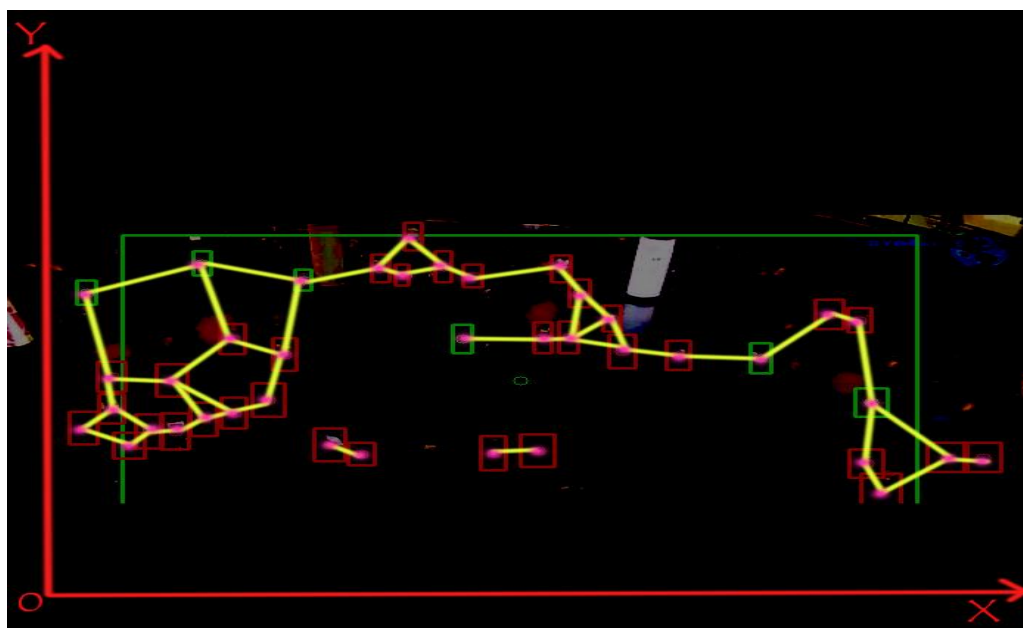
Frame:1, Step-1. Total Person: 43

Table 4.33 **Person Detection**

Frame-1, Step-2. Person Detected: 42

Table 4.34

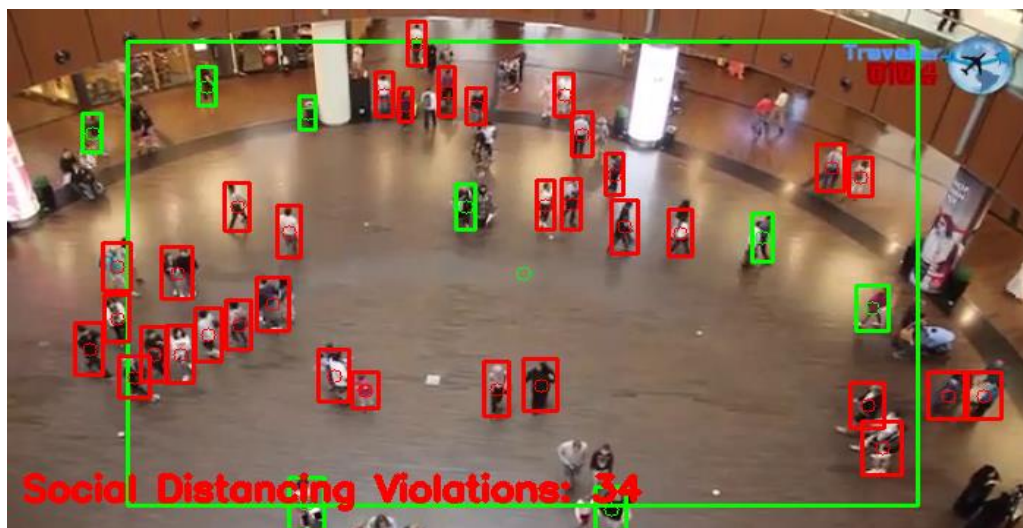
Person detection and Euclidean distance measurement



Frame: 1, Step-3. Applying Euclidean Distance Formula

Table 4.35

Social Distance Violence estimated



Frame: 1, Step-4. Detected Person: 42

Table 4.36 More Sample Frames from the same video

 Frame:2 Total Person: 43 Correctly Detected: 42	 Frame:3 Total Person: 43 Correctly Detected: 42
 Frame:4 Total Person: 43 Correctly Detected: 42	 Frame:5 Total Person: 43 Correctly Detected: 42

Here,

Tracking Accuracy for frame in percentage = $\frac{\text{No. of Correctly Detected} \times 100}{\text{No. of Total person}}$

Frame 1= 98% , Frame 2= 98%, Frame 3= 98% , Frame 4= 98% , Frame 5= 98%

Tracking Accuracy of whole environment in percentage = $\frac{\text{The Sum of Tracking Accuracy for frames in percentage}}{\text{Total Frame}} = \frac{98+98+98+98+98}{5} = 98\%$
(approximate)

Tracking Accuracy gained from Crowded Footage: 98%

Table 4.37 Tracking Accuracy in % is shown in the basis of The Total Detected Object (person) and Correctly Detected Object (Person)

ENVIRONMENT NAMES	FRAME NO.	NO. OF TOTAL PERSON	NO. OF CORRECTLY DETECTED	TRACKING ACCURACY FOR FRAME IN PERCENTAGE (NO. OF CORRECTLY DETECTED*100 / NO. OF TOTAL PERSON) (APPROXIMATE)	TRACKING ACCURACY OF WHOLE ENVIRONMENT IN PERCENTAGE = SUM OF TRACKING ACCURACY FOR FRAME IN PERCENTAGE /TOTAL FRAME (APPROXIMATE)
GOOD QUALITY	FRAME : 1	45	45	100%	99%
	FRAME : 2	40	39	98%	
	FRAME : 3	40	39	98%	
	FRAME : 4	40	39	98%	
	FRAME : 5	38	38	100%	
FUZZY	FRAME : 1	07	05	63%	64%
	FRAME : 2	08	05	63%	

	FRAME : 3	08	05	63%	
	FRAME : 4	07	04	57%	
	FRAME : 5	08	05	72%	
DAYLIGHT	FRAME : 1	16	14	88%	87%
	FRAME : 2	12	10	84%	
	FRAME : 3	13	12	93%	
	FRAME : 4	13	11	85%	
	FRAME : 5	12	10	84%	
NIGHTLIGHT	FRAME : 1	05	05	100%	92%
	FRAME : 2	05	04	80%	

	FRAME : 3	05	04	80%	
	FRAME : 4	05	05	100%	
	FRAME : 5	05	05	100%	
LESS POPULATED	FRAME : 1	03	03	100%	100%
	FRAME : 2	03	03	100%	
	FRAME : 3	03	03	100%	
	FRAME : 4	03	03	100%	
	FRAME : 5	03	03	100%	
CROWDED	FRAME : 1	43	42	98%	98%
	FRAME : 2	43	42	98%	

	FRAME : 3	43	42	98%	
	FRAME : 4	43	42	98%	
	FRAME : 5	43	42	98%	
					TRACKING ACCURACY AVERAGE = 90% (APPROXIMATE)

Here, Sample Video = 6

Total Frames = 30

- 1) Frame 1: Tracking Accuracy gained from Good Quality Footage: 99%
- 2) Frame 2: Tracking Accuracy gained from Fuzzy Footage: 64%
- 3) Frame 3: Tracking Accuracy gained from Daylight Footage: 87%
- 4) Frame 4: Tracking Accuracy gained from Nightlight Fuzzy Footage: 92%
- 5) Frame 5: Tracking Accuracy gained from Less Populated Footage: 100%
- 6) Frame 6: Tracking Accuracy gained from Crowded Footage: 98%

Average Tracking Accuracy gained from six different environment = 90% (approximate)

The proposed method estimate social distance from high quality footage 99% and in fuzzy footage it's also quite good in percentage. Six types of top-down videos from different places are used as samples. Here, Total person and Correctly Detected Object (Person) are calculated to get the percentage of accuracy. As a result, 90% of tracking accuracy is found which is quite good as social distance estimation from top-down footages.

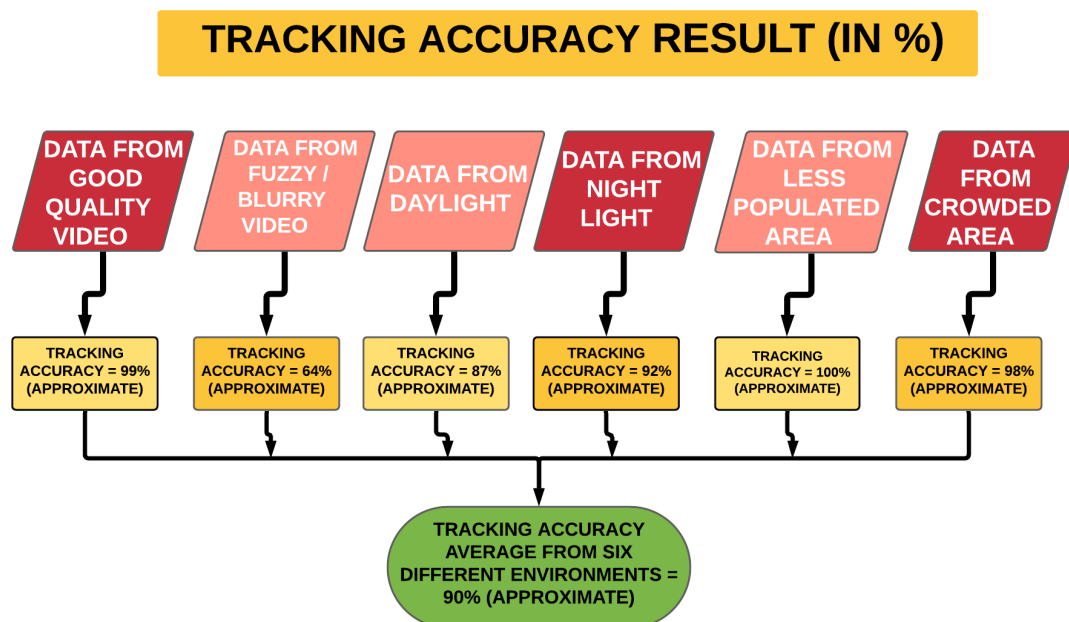


Figure 4.2 Tracking Accuracy Result (in %)

4.3 EXPERIMENTAL SETUP

4.3.1 Hardware & Software requirement:

The tools are used in this proposed system are listed below:

- CUDA-capable GPU setup
- Python3.8

- Spyder(Scientific Python Development Environment)
- Google Colaboratory

4.3.2 Libraries

- OpenCv Python==4.2.0.34
- Numpy==1.18.5
- Imutils==0.5.3
- Matplotlib
- Argparse
- Scipy==1.4.1

4.4 SUMMARY

This chapter shows the result of the proposed method with frames and necessary figures. Also the way Euclidean distance formula works are also shown. Frames from video samples proves the real life implementation and the table shown gave the aspired accuracy implementing the latest model.

CHAPTER V

CONCLUSION AND FUTURE WORKS

5.1 CONCLUSION

This research has been directed a precise review where the new advancement of deep learning networks object recognition is obvious, deep models have essentially improved execution, however there are as yet numerous challenges and difficulties. With YOLO in a scientific python development environment we got 92% of tracking accuracy which proves it's tremendous performance as a Social Distance Detection system.

5.2 CONTRIBUTION OF THIS THESIS

The major contributions in the thesis is: Social Distance Estimation for top-down CCTV footage using the latest object detection model of Convolutional Neural Network YOLO and Euclidean distance in six different environments and gaining a tracking accuracy average from them.

5.3 FUTURE WORKS

Some aspects of the present research work can be further investigated and improved. Based on the literature reviews and studies conducted in this thesis, the following recommendation is proposed: The proposed method mapped distances for top-down videos or CCTV Footages. So, Future works can be with side-to-side footages and also with the updated versions of object detection models if any.

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